



The Gainward GeForce FX 5200 Ultra performed the best in the low-end category

leaving us wondering where the rest of the memory went. The most probable reason for this would be that both the 64 MB and the 128 MB versions have the same number of memory chips, thereby giving the 64 MB version much more parallel access than the 128 MB one. While this might make the 64 MB card look

like a better option, this need not be the case—it can only be ascertained after more heavy duty games such as *DOOM 3*, which make use of massive texture mapping, are released. The 64 MB version will then have to kick in the AGP bus to move textures to the main RAM on the system, whereas the 128 MB one will have the advantage of processing them on the card itself.

In the 5200 Ultra category, we only had the Gainward 5200 Ultra video card which does what it is meant to do—provide a good performance-to-price ratio. It will play next generation games such as *Half-Life 2* and *Doom3*, albeit at a low resolution. The DX9 benchmark saw the normal 5200s lag behind, due to the stripped down clock and memory speeds which this card covers up. In fact, this card's performance was almost equal to that of a non-ultra GeForce FX 5600, which is very good, considering its price. All in all, this card should provide the value user a good gaming experience for some time to come.

Mid-range

This range consisted of the 5600 and the 9600 series graphics cards. This is a slightly more demanding market segment, which needs lots of performance at a decent price. Here price-to-performance is an important factor.

This is currently the most competitive market—both ATi and nVidia have excellent offerings in this range. The Radeon 9600 and its PRO version are the primary offerings from ATi. Both these give very good performance, but the pricing of the 9600 PRO is slightly higher than the 5600 series by nVidia.

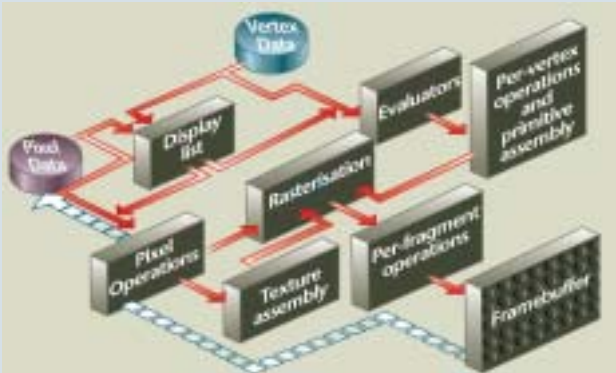
The 9600 PRO is a scaled-down version of the 9800 PRO. It is currently aimed at replacing the 9700, and the 9500 series, which hasn't yet been released in India. The card's performance is lower than that of both the 9700 and the 9600 PRO. It is code-named

Info TransGaming has come up with a Linux port of DirectX called WineX. Currently at version 3.1, it has support for DirectX 8. Some of the games they have ported using WineX are: *Diablo*, *Grand Theft Auto: Vice City*, *SimCity*, *Warcraft* and *The Sims*



OpenGL Technology

OpenGL has a rendering pipeline that directs different elements of the scene via different processes, to the framebuffer. The OpenGL standard differentiates between two types of data—Geometric and Pixel. Geometric data resembles geometric shapes (vertices, line and polygons), whereas pixel data consists of pixels, bitmaps and images. Geometric data are passed through Evaluators, and various other pre-vertex operations, before they are rasterised. Meanwhile, the pixel data are passed through a separate branch of the pipeline that consists of various pixel operations. Finally, both the data are passed to the rasterising phase, at which point they are combined and converted into fragments. Just before rasterisation, various textures are applied to the pixel and geometric data to give them a realistic appearance. Finally, various per-fragment operations are carried out on the rasterised data, which are then stored in the framebuffer to be sent out to the display unit.



The OpenGL rendering pipeline divides raw data into pixel and vertex data, and then processes them separately

the RV350. The Radeon 9800, 9200 and the 9600 were all announced at the same time by ATi; however, shipping was delayed, which caused some noise in the market. The 9600 is an important achievement for ATi, built as it is on the 0.13 micron fabrication process, which will decrease in the overall chip manufacturing cost.

Before we go into the tests and their results, a little background is necessary in order to understand the difference between the 9800 PRO and the 9600 PRO. The 9600 is the RV350 chipset based on the R350 chipset of the 9800 PRO. The RV350 features four full DX9 pixel pipelines and two ver-

tex shaders. This is exactly the R350 halved. The two variants of the RV350 are the 9600 and the 9600 PRO running at 325/400 (200 MHz DDR) and 400/600 (300 MHz DDR) respectively. The overall effect of this is that ATi is expecting that the performance-to-price ratio of the



The Compro Poladyne 5600: A great combination of price and performance